

MIT805 – Group Project – Part 1

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Data Collection and Data Analysis

The yellow trip dataset is collected by technology vendors who are authorized under the Taxicab & Livery Passenger Enhancement Programs (TPEP/LPEP). They provide the data to the NYC (New York City) Taxi and Limousine Commission (TLC) who publish it monthly on their website with a two month delay. The datasets includes valuable parameters to better understand the Taxi operations in NYC. It is licensed by the City of New York Terms of Use. It is academically appropriate for use because it is publicly released government open data and available for download and analysis.

The study uses the NYC yellow Taxi trip data, which is provided in Parquet format to support efficient processing and with 20 data fields. The raw NYC yellow trip dataset has a total size of approximately 29.96GB, covering trips from January 2009 until May 2026. To ensure that the working dataset provides a relevant representation of current tax-trip patterns, the most recent available data was prioritized. Starting from 2026 and moving backwards chronologically, the cumulative dataset size was calculated until the required working dataset (greater than 12GB) and the processing dataset (greater than 3GB) was reached, refer to **Table 1**. The working dataset comprises of data from January 2014 until May 2026, while the processing dataset comprises of data from January 2021 until May 2026.

As the dataset consisted of monthly parquet files, there was inconsistencies in the column name and data types across the files. Column names were therefore standardized to ensure consistency when combining the monthly datasets. Differences were also identified for several variables, for example, airport fee and passenger count were represented as both integer and double types. These inconsistencies were resolved by assigning a common data type that preserved the information contained in the source data. Financial and measurement variables were standardized to a double data type while identifier variable were retained as integer types were appropriate.

Null values were also identified and replaced according to the nature of each variable. Missing passenger count which was 2.64% of the total count was replaced with the mode, missing rate code ID values with 99 which is the code for unknown as per documentation, and the missing store and forward flag values with "Unknown". Missing financial variables were replaced with zero. Individual variables were examined for invalid or anomalous records. A subset of trips contains pick up and drop off datetimes corresponding to 2001 and 2002. These records accounted for 4.86% of the dataset with no identifiable pattern or valid explanation, and were therefore removed. Negative financial values were retained as they were associated with disputed or refunded transactions, while negative trip distance values representing 0.01% of the data, were removed as invalid. Vendor ID not listed in the TLC data dictionary were retained due to insufficient information to determine their validity.

The temporal analysis shows clear change in taxi demands across years, months and hours. Annual trip volume declined sharply from 2019 to 2020 as shown in **Figure 1**, coinciding with the COVID-19 pandemic and the resulting restriction on travel and movements. Although demand recovered after 2020, it remained below pre-pandemic levels, indicating a longer term change in taxi usage. Monthly demand also shows a recurring decline during July and August as shown in **Figure 2**, which may reflect seasonal changes in travel patterns during the summer period. The hourly analysis shows demand is lowest during the early morning hours and increases from midday, with the highest concentration occurring between 17:00 and 19:00 peaking at around 18:00, refer to **Figure 3**. This pattern is notable because demand is not concentrated during the early morning period as might be expected from traditional patterns. These patterns can support taxi operators in aligning driver availability and fleet deployment with periods of higher demand, potentially improving vehicle utilization and reducing idle time during low-demand periods.

The relationship between trip distance and fare amount shows a positive association, with longer trips generating higher average fares, as shown in **Figure 4**. Similarly, for credit-card transactions, average tip amount increased with trip distance depicted in **Figure 5**, indicating that longer trips may provide drivers with higher fare and tipping opportunities. Differences in fare value were also observed across rate codes. Rate Code 4 (Nassau or Westchester) recorded the highest average fare, followed by Rate code 3 (Newark) and rate code 5 (negotiated fare), depicted in **Figure 6**. These rate codes represent trips involving destinations outside the standard New York City rate structure ; for example rate code 4 applies to trips to Nassau and Westchester, while rate code 3 applies to Newark Airport. This suggests an opportunity for taxi operators to prioritize or strategically position vehicles for higher-value trip segments where operationally feasible.

The data analysis on NYC yellow taxi dataset demonstrates volume through the large number of taxi trips analyzed and velocity through changes in demand across years, months and hours. Variability is observed in seasonal and hourly demand patterns, as well as differences in fare and tip amounts across trip distances and rate codes. Variety is reflected in the range of temporal, geographic, trip, payment and financial variables available in the dataset. Veracity was addressed through data cleaning, standardization and validation of the dataset. The analysis provides Value by identifying periods of high demand and higher-value trip segments that can support improved driver and fleet allocation. Finally, Visualization was used to communicate these patterns and make the identified trends and opportunities easier to interpret. The dataset contains some inconsistencies and anomalous records discussed. It also lacks contextual variables such as weather, traffic and public transport conditions that may influence demand .

References

New York City Taxi and Limousine Commission. (n.d.). TLC trip record data. [TLC Trip Record Data - TLC](#).

GitHub repo link

[ntomben97/NYC-Yellow-Taxi-Big-Data-Project](#)

Tables

Table 1: NYC Yellow data sizes by year.

Year	Size (GB)	Cumulative Size (GB)
2026	0.302919	0.302919
2025	0.772973	1.075892
2024	0.645408	1.721300
2023	0.592082	2.313383
2022	0.572856	2.886239
2021	0.446644	3.332883
2020	0.348873	3.681756
2019	1.158130	4.839886
2018	1.363844	6.203731
2017	1.485167	7.688898
2016	1.707174	9.396073
2015	1.887845	11.283918
2014	2.000711	13.284628
2013	1.996676	15.281304
2012	2.033804	17.315108
2011	2.056566	19.371674
2010	5.266181	24.637855
2009	5.324814	29.962669

Figures:

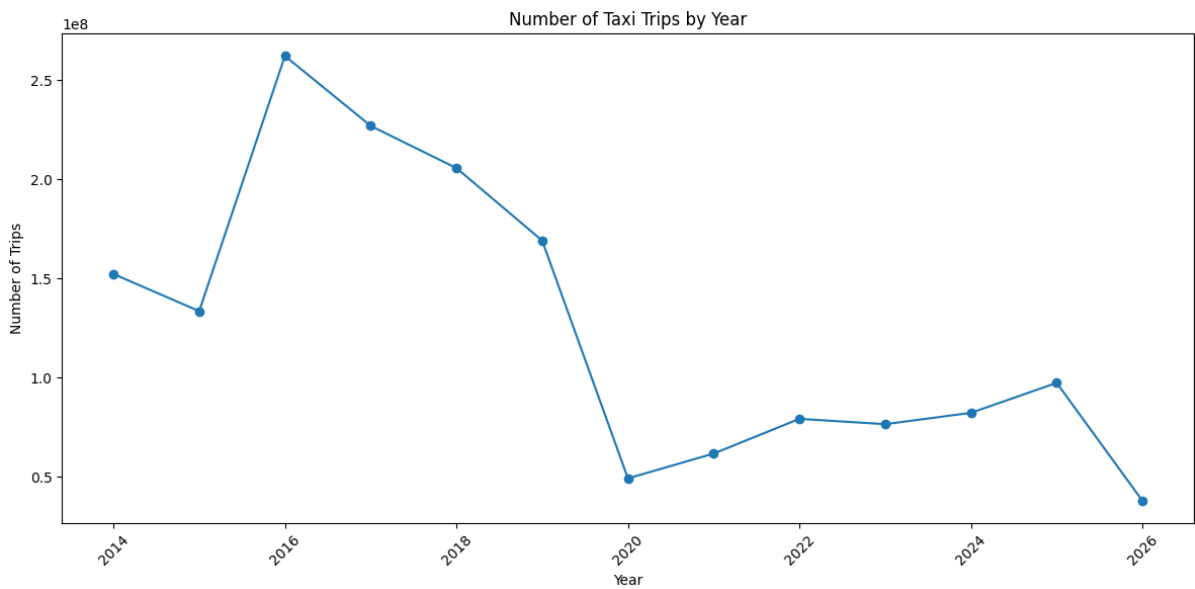


Figure 1: Number of Taxi Trips by Year.

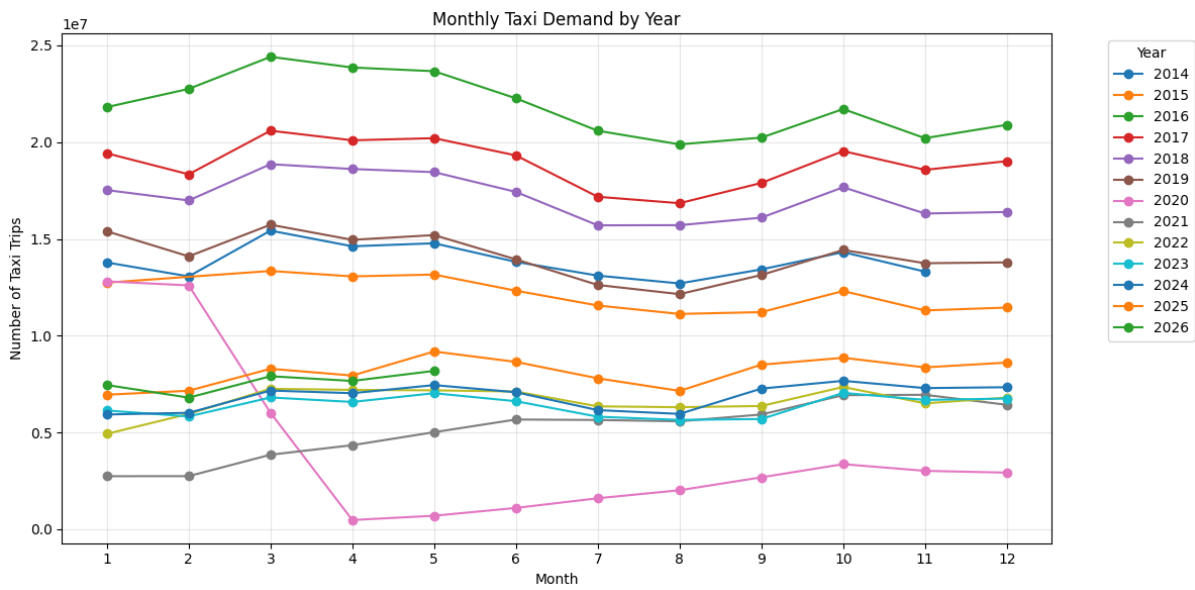


Figure 2: Monthly Taxi Demand by Year.

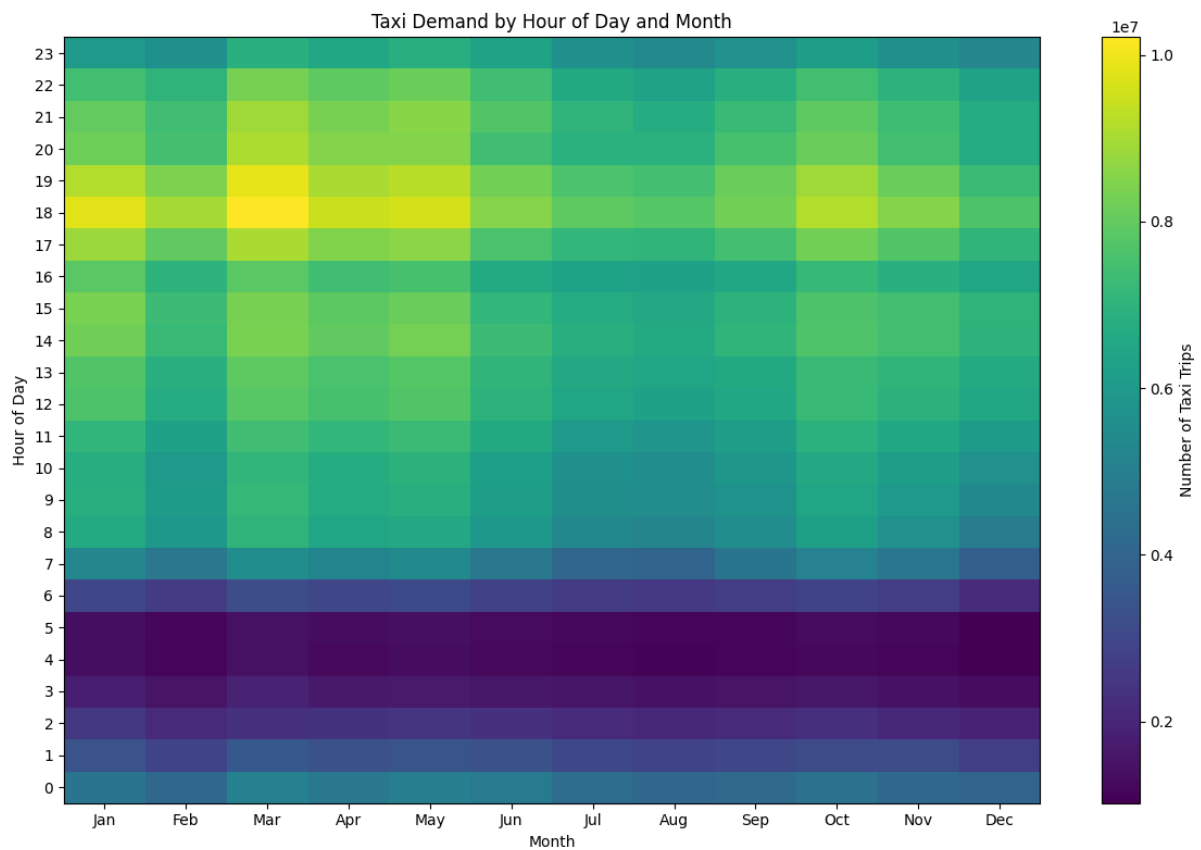


Figure 3: Taxi demand by hour and month.

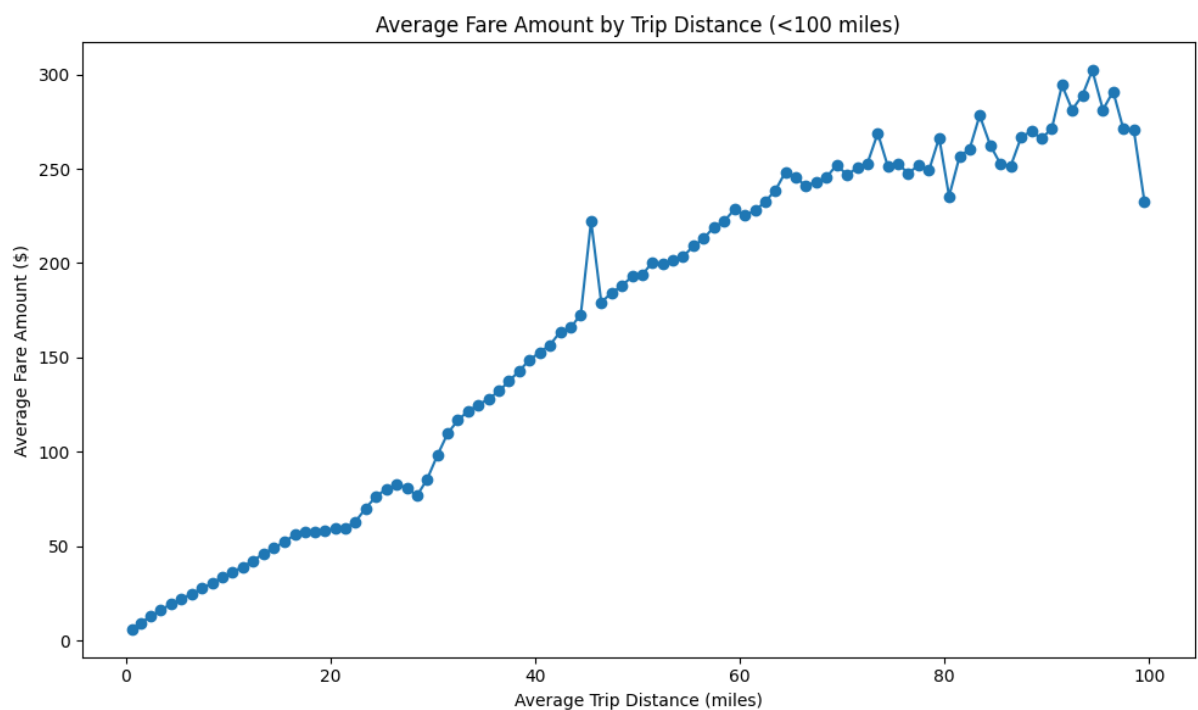


Figure 4: Average Fare Amount by Trip Distance.

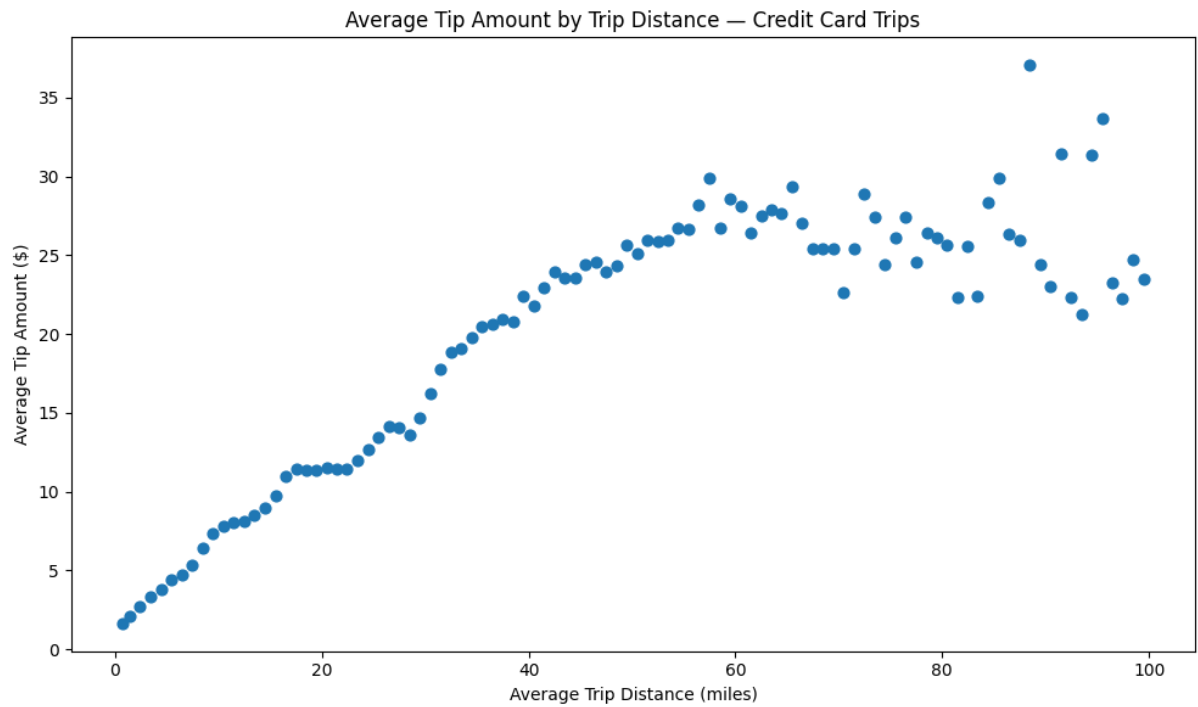


Figure 5: Average Tip Amount by Trip Distance.

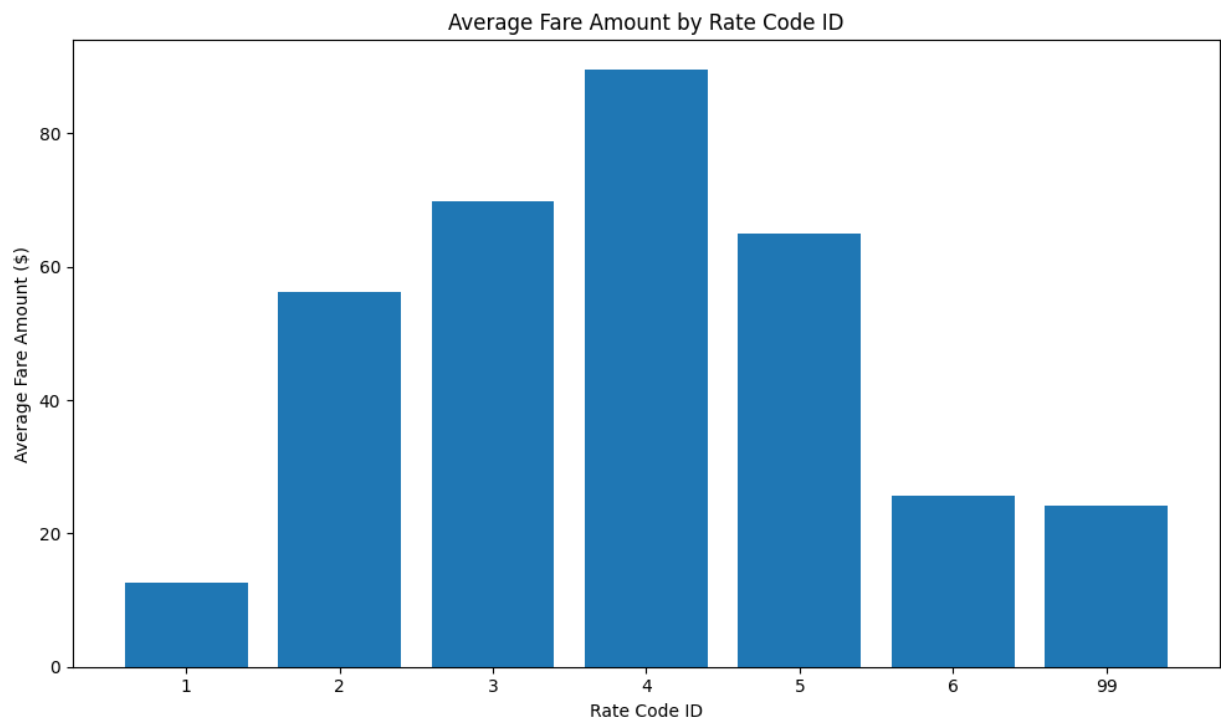


Figure 6: Average Fare Amount by Rate Code ID.